W9 HARD

SCENARIO PRACTICE

Connecting Smart Objects - Exam Practice

6LoWPAN | LOADng | RPL | 802.11ah | NB-PLC | RFID

WHAT THIS PDF TRAINS

- Long 8-mark scenario questions
- Technology identification from clues
- Hard comparison and design questions
- Routing failure and troubleshooting
- Short traps and rapid revision
- Exam answer formula + memory triggers

Top Priority

Practice first: 6LoWPAN -> LOADng -> RPL -> 802.11ah -> RFID

Then: NB-PLC -> 802.15.4e/g -> ZigBee IP/MAC

Answer pattern: Identify -> Reason -> 5-7 points -> Conclusion

Scenario 1: Smart Agriculture + IPv6

Long-question pattern - identify, justify, explain

SCENARIO

A farm has hundreds of tiny battery-powered sensors using IEEE 802.15.4 radios. Each sensor has limited processing ability, but the owner wants every device to use IPv6 and connect to the Internet.

QUESTION

Which technology should be used? Explain why it fits this constrained IoT situation.

Best answer: 6LoWPAN

- Designed for low-power wireless personal area networks.
- Allows small devices with limited processing ability to use Internet Protocol.
- Enables low-power devices to connect to the Internet.
- Allows IEEE 802.15.4 radios to carry 128-bit IPv6 addresses.
- Uses header compression and address translation techniques.
- Compresses and reformats IPv6 packets to fit IEEE 802.15.4 packet format.
- Common uses include IoT, Smart Grid and M2M.

MEMORY

small device + 802.15.4 + IPv6 = 6LoWPAN

Scenario 2: Large IPv6 Packet Problem

Long-question pattern - identify, justify, explain

SCENARIO

An engineer must send IPv6 traffic through a small IEEE 802.15.4 frame. The original packet does not fit directly and may need to be split while still preserving routing information.

QUESTION

Which 6LoWPAN mechanisms and header types are relevant?

Best answer: 6LoWPAN Adaptation + DMF headers

- 6LoWPAN compresses and reformats IPv6 packets.
- Dispatch Header identifies the next header type and initiates communication.
- Mesh Addressing Header carries mesh-routing information such as Hops Left.
- Hops Left is decremented by each forwarding node.
- Fragmentation Header supports first and subsequent fragments.
- Fragmentation fields include datagram size, tag, and offset.

MEMORY

DMF = Dispatch - Mesh - Fragmentation

Scenario 3: Global vs Local Address

Long-question pattern - identify, justify, explain

SCENARIO

Inside a 6LoWPAN PAN, Device A needs a globally unique identifier. Device B only needs a short local address for use within the PAN.

QUESTION

Which address should each device use and why?

Best answer: 64-bit for global, 16-bit for PAN-specific

- 64-bit extended addresses are globally unique.
- 16-bit short addresses are PAN specific.
- The PAN coordinator assigns the 16-bit short address.
- IPv6 multicast is not directly supported by IEEE 802.15.4.
- IPv6 packets can be carried as link-layer broadcast frames.

MEMORY

64 = Global | 16 = Local PAN

Scenario 4: On-Demand Route Discovery

Long-question pattern - identify, justify, explain

SCENARIO

A low-power IoT router has no route to a destination. It generates a route request, neighboring routers forward it, and the destination sends a route reply back hop by hop.

QUESTION

Identify the routing protocol and explain the process.

Best answer: LOADng

- LOADng is derived from AODV and extended for IoT.
- Originator generates a Route Request (RREQ).
- Routers forward the RREQ until it reaches the destination.
- Destination generates a Route Reply (RREP).
- RREP is forwarded back by unicast hop-by-hop toward the originator.
- The protocol is based on on-demand route discovery.

MEMORY

RREQ -> RREP = Request -> Reply

Scenario 5: Broken LOADng Route

Long-question pattern - identify, justify, explain

SCENARIO

A sensor is sending data over a valid LOADng route. One link suddenly fails and the sender must be informed that the route is no longer usable.

QUESTION

How does LOADng report the failure? Mention other important rules.

Best answer: RERR + destination-only reply

- A Route Error (RERR) is returned to the originator when a route break is detected.
- LOADng supports optimized flooding to reduce RREQ overhead.
- Only the destination is allowed to respond to an RREQ.
- Intermediate LOADng routers cannot respond even if they know a valid route.
- RREQ/RREP messages from a router share a unique monotonically increasing sequence number.

MEMORY

Request -> Reply -> Error = RREQ -> RREP -> RERR

Scenario 6: Low-Power Lossy Network

Long-question pattern - identify, justify, explain

SCENARIO

A smart-city network has low-power nodes and unreliable wireless links. The routing topology should be maintained continuously, but the protocol should also react when a node or link goes down.

QUESTION

Which protocol is suitable? Explain the clues.

Best answer: RPL

- RPL is a Distance Vector IPv6 routing protocol for Low-Power and Lossy Networks (LLN).
- Maintains routing topology using low-rate beaconing.
- Beaconing rate increases when inconsistencies are detected.
- Routing information can be included in the datagram.
- Proactive behavior maintains topology.
- Reactive behavior resolves routing inconsistencies.

MEMORY

LLN + beacon + proactive/reactive = RPL

Scenario 7: Energy and Loop Optimization

Long-question pattern - identify, justify, explain

SCENARIO

An IPv6 routing protocol must work in a constrained sensor network. It should minimize energy and latency, consider node power and bandwidth, and detect routing loops.

QUESTION

Which protocol provides these features?

Best answer: RPL

- RPL supports message confidentiality and integrity.
- Supports Data-Path Validation.
- Supports Loop Detection.
- Routing objectives can minimize energy.
- Routing objectives can minimize latency.
- Can satisfy constraints involving node power, bandwidth and other limits.
- RPL requires bidirectional links and verifies parent reachability.

MEMORY

RPL = secure + save Energy/Latency + detect Loops

Scenario 8: Massive Smart Meter Wi-Fi

Long-question pattern - identify, justify, explain

SCENARIO

A utility company wants one Wi-Fi access point to serve thousands of low-power smart meters over longer range. Normal Wi-Fi is not ideal for sub-GHz coverage, battery life, and very large device counts.

QUESTION

Which IEEE technology is designed for this?

Best answer: IEEE 802.11ah / Wi-Fi HaLow

- Provides a sub-GHz version of Wi-Fi.
- Designed for IoT sensors, meters and extended-range applications.
- Supports up to 8192 devices per access point.
- Uses a shortened MAC header.
- Supports NDP (Null Data Packet).
- Uses grouping and sectorization to reduce contention.
- Includes RAW, TWT, and Speed Frame Exchange mechanisms.

MEMORY

8192 + sub-GHz + IoT Wi-Fi = IEEE 802.11ah

Scenario 9: Battery-Saving Wake Schedule

Long-question pattern - identify, justify, explain

SCENARIO

A battery-powered IEEE 802.11ah sensor should sleep most of the time. The access point decides exactly when the device can wake and access the network.

QUESTION

Which feature is being used and what problem does it solve?

Best answer: TWT - Target Wake Time

- Access point defines when a device can access the network.
- Device can remain in a low-power state until its TWT arrives.
- Reduces energy consumption.
- Reduces collision probability in large cells with many clients.
- Supports better battery life for constrained devices.

MEMORY

TWT = Time to Wake

Scenario 10: Too Many Devices Transmitting

Long-question pattern - identify, justify, explain

SCENARIO

A large IEEE 802.11ah deployment suffers collisions because many devices attempt to transmit at the same time. The system needs fair and controlled medium access.

QUESTION

Which 802.11ah feature addresses this problem?

Best answer: RAW - Restricted Access Window

- Avoids simultaneous transmissions when many devices are present.
- Provides fair access to the wireless network.
- Improves medium-access efficiency.
- Reduces collisions.
- Can also save power for battery-powered devices.

MEMORY

RAW = Restrict Access

Scenario 11: Power-Line Smart Metering

Long-question pattern - identify, justify, explain

SCENARIO

A utility provider wants meters and grid devices to communicate without relying on wireless radio. The same electrical wires used to carry power should also carry IoT data.

QUESTION

Which technology should be selected? Give applications.

Best answer: IEEE 1901.2a / NB-PLC

- IEEE 1901.2a is a wired technology.
- Standard for Narrowband Power Line Communication (NB-PLC).
- Uses the same wires that carry electrical power.
- Supports low power, long range, and resistance to interference.
- Applications include smart metering and distribution automation.
- Also used for public lighting, EV charging stations, microgrids, and renewable energy.

MEMORY

electric wire + data = NB-PLC

Scenario 12: Industrial MAC Problems

Long-question pattern - identify, justify, explain

SCENARIO

An IEEE 802.15.4 network in a factory experiences MAC reliability problems, unbounded latency and multipath fading. The system also supports factory automation.

QUESTION

Which IEEE amendment targets these issues?

Best answer: IEEE 802.15.4e

- Expands the MAC layer feature set.
- Improves MAC reliability.
- Addresses unbounded latency.
- Addresses multipath fading.
- Improves support for factory and process automation.
- Also supports Smart Grid application domains.

MEMORY

E = Enhance MAC

Scenario 13: Smart Utility Network

Long-question pattern - identify, justify, explain

SCENARIO

A Smart Grid project uses the term Wi-SUN and focuses on Smart Utility Network communication.

QUESTION

Which IEEE 802.15.4 amendment is most closely associated with this use?

Best answer: IEEE 802.15.4g

- IEEE 802.15.4g is associated with Smart Utility Networks.
- Wi-SUN appears as a standardization/alliance term in this context.
- SUN stands for Smart Utility Network.
- The technology is important for utility and Smart Grid applications.

MEMORY

G = Grid

Scenario 14: RFID Warehouse Tracking

Long-question pattern - identify, justify, explain

SCENARIO

A warehouse wants to identify and track boxes automatically. Labels should be readable without direct visual alignment or line-of-sight, and data should be stored in the company database.

QUESTION

Why is RFID suitable?

Best answer: RFID

- RFID means Radio-Frequency Identification.
- Data is digitally encoded in RFID tags and read by a reader.
- Tag data can be stored in a database.
- RFID can be read outside direct line-of-sight, unlike traditional barcode/QR scanning.
- Useful for inventory management, asset tracking and supply-chain management.

MEMORY

no line-of-sight + tag/reader = RFID

Scenario 15: Passive vs Active RFID

Long-question pattern - identify, justify, explain

SCENARIO

Tag A has no battery and only transmits after receiving energy from the RFID reader. Tag B contains its own power source.

QUESTION

Identify Tag A and Tag B and explain the difference.

Best answer: A = Passive RFID, B = Active RFID

- Passive RFID tags are the most widely used.
- Passive tags must be powered inductively by the reader before transmitting.
- Active RFID tags have their own power supply.
- An RFID tag includes an integrated circuit and an antenna.
- Protective material can shield the tag from environmental effects.

MEMORY

Passive = Power Please | Active = Already Powered

Scenario 16: RFID Working Principle

Long-question pattern - identify, justify, explain

SCENARIO

A company wants automatic object identification with little or no human intervention. A radio reader communicates with tags and the captured data is transferred to a computer system.

QUESTION

Explain the working principle and system components.

Best answer: RFID based on AIDC

- RFID is derived from Automatic Identification and Data Capture (AIDC).
- AIDC performs object identification, data collection and mapping to computer systems.
- RFID uses radio waves to perform AIDC functions.
- Main components are RFID tag/smart label, RFID reader, and antenna.
- The reader collects tag information and passes it to the computer/database system.

MEMORY

TRA = Tag - Reader - Antenna

Scenario 17: Full Smart-City Design

Hard integrated question - combine multiple Week 9 technologies

HARD DESIGN

A smart city needs thousands of smart meters, some devices communicating over power lines, low-power IPv6 sensors, routing in lossy networks, and RFID-based asset tracking.

Design a Week 9 solution:

Smart meters / long-range Wi-Fi	->	IEEE 802.11ah / Wi-Fi HaLow
Power-line devices	->	IEEE 1901.2a / NB-PLC
Low-power IPv6 sensors	->	6LoWPAN
Lossy IPv6 routing	->	RPL
On-demand route discovery	->	LOADng
Asset / object identification	->	RFID

MASTER DESIGN MEMORY

Internet = 6LoWPAN | Route-on-demand = LOADng | LLN = RPL | Long-range Wi-Fi = 802.11ah | Power wire = NB-PLC | Identify = RFID

Hard Trap Questions

True/False + one-line correction

1. **6LoWPAN is only for high-power computers.**

Ans: FALSE - it is for low-power constrained devices.

2. **64-bit 6LoWPAN addresses are PAN specific.**

Ans: FALSE - 64-bit addresses are globally unique.

3. **Intermediate LOADng routers may answer any RREQ.**

Ans: FALSE - only the destination may answer.

4. **RPL is designed for Low-Power and Lossy Networks.**

Ans: TRUE.

5. **RAW mainly schedules sleeping time.**

Ans: FALSE - TWT schedules wake time; RAW controls access.

6. **IEEE 1901.2a is a wireless standard.**

Ans: FALSE - it is wired NB-PLC.

7. **Passive RFID tags have their own battery.**

Ans: FALSE - passive tags are powered by the reader.

8. **RFID requires direct line-of-sight.**

Ans: FALSE - RFID can be read outside line-of-sight.

Rapid Practice - 30 Short Questions

Try without looking at the notes

1. What does 6LoWPAN allow?
2. What kind of address does IPv6 use?
3. Which 6LoWPAN address is globally unique?
4. Who assigns the 16-bit short address?
5. Name three 6LoWPAN header types.
6. What happens to Hops Left at each node?
7. Name two 6LoWPAN routing protocols.
8. LOADng is derived from which protocol?
9. What is RREQ?
10. What is RREP?
11. What is RERR?
12. Which protocol uses low-rate beaconing?
13. What does LLN mean?
14. Is RPL proactive, reactive, or both?
15. Name two RPL security/validation features.
16. What does RAW do?
17. What does TWT do?
18. Maximum devices per 802.11ah access point?
19. What is Wi-Fi HaLow?
20. What is NDP?
21. What is NB-PLC?
22. Is IEEE 1901.2a wired or wireless?
23. What does RFID stand for?
24. Does RFID require line-of-sight?
25. What powers a passive RFID tag?
26. What powers an active RFID tag?
27. Name RFID system components.
28. Name three RFID applications.
29. What problem does IEEE 802.15.4e improve?
30. What does Wi-SUN stand for?

Final Exam Strategy

How to identify unseen Week 9 scenarios quickly

MASTER DECISION MAP

IPv6 + low-power 802.15.4	->	6LoWPAN
RREQ / RREP / RERR	->	LOADng
LLN + beacon + proactive/reactive	->	RPL
8192 + sub-GHz + RAW/TWT	->	IEEE 802.11ah
Electrical wire + IoT data	->	IEEE 1901.2a
MAC reliability / latency / fading	->	IEEE 802.15.4e
Smart Utility / Wi-SUN	->	IEEE 802.15.4g
Tag + reader + no line-of-sight	->	RFID

8-MARK ANSWER FORMULA

- 1. Identify the technology immediately.
- 2. Write a one-line definition.
- 3. Connect the scenario clues to the technology.
- 4. Write 5-7 syllabus points.
- 5. Add one exact keyword, number, or protocol sequence.
- 6. Finish with a clear conclusion.

TONIGHT'S TOP 8

6LoWPAN | LOADng Route Discovery | LOADng RERR | RPL | 802.11ah | RAW vs TWT | NB-PLC | RFID Passive/Active